作业

作业来自

梁昆淼《力学》 第 4 版 高等教育出版社

Johns Analytical Mechanics 2nd ed Oxford University Press

例如： L01.08 表示梁昆淼书第 01 章第 08 题； J02.05 Johns 书第 02 章第 05 题

J03.07† 表示 Johns 书第 03 章第 07 题，带 † 表示仅要求荣誉班同学完成

Homework 03, Week 04 2023.09.25 Due 2023.10.09 J02.05, J02.06, J02.09, J02.12

★ J02.05

Suppose that two masses have a motion defined by a Lagrangian function

$$L = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - U(|\vec{r}_1 - \vec{r}_2|)$$

where $\vec{r}_1$ and $\vec{r}_2$ denote the position vectors of masses 1 and 2, respectivley,

$$\vec{v}_1 = \frac{d\vec{r}_1}{dt}, \quad \vec{v}_2 = \frac{d\vec{r}_2}{dt}, \quad v_1^2 = \vec{v}_1 \cdot \vec{v}_1, \quad v_2^2 = \vec{v}_2 \cdot \vec{v}_2.$$

Let $\vec{r} = \vec{r}_2 - \vec{r}_1, \quad \vec{R} = \dfrac{m_2 \vec{r}_2 + m_1 \vec{r}_1}{m_1 + m_2}$

(a) Rewrite the Lagrangian in terms of the variables $\vec{r}$ and $\vec{R}$ and their time derivatives.

Show the Lagrangian can be written as two terms, one of which depends only on $\vec{R}$ and its time derivative and the other only on $\vec{r}$ and its time derivative. Such Lagrangian system are called separable.

(b) Show that three components of $\vec{R}$ are ignorable coordinates and that the total momentum of the system is conserved.

Solution:

★ J02.06

A mass m is acted on by a force derived from the generalized potential

$U^{(\text{vel})}(\vec{r}, \vec{v}, t) = V(r) + \vec{\sigma} \cdot \vec{L}$ with the superscript "(vel)" denoting velocity dependent potential

where $r = \sqrt{x^2 + y^2 + z^2}$, $\vec{\sigma} = \sigma \vec{e}_z$, $\vec{L} = \vec{r} \times m \vec{v}$ and $\vec{r}$ and $\vec{v}$ are the position and velocity of the mass relative to some inertial coordinate system.

(a) Express $U^{(\text{vel})}$ in Cartesian coordinates (the s system) $x, y, z, \dot{x}, \dot{y}, \dot{z}$ and find the force $\vec{F}$ and its three Cartesian components F_x, F_y, F_z

(b) Express $U^{(\text{vel})}$ in spherical polar coordinates (that can be called the q system) $r, \theta, \phi, \dot{r}, \dot{\theta}, \dot{\phi}$ and find the generalized force Q_k, $k = 1, 2, 3$

(c) The force vector found in (a) can be rewritten in terms of spherical polar unit vectors as

$\vec{F} = F_r \vec{e}_r + F_\theta \vec{e}_\theta + F_\phi \vec{e}_\phi$ with $F_r = \vec{e}_r \cdot \vec{F}$, $F_\theta = \vec{e}_\theta \cdot \vec{F}$, $F_\phi = \vec{e}_\phi \cdot \vec{F}$. Show $Q_r = F_r$.

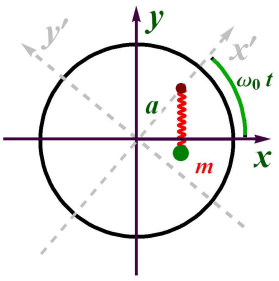
(d) However $Q_\phi \neq F_\phi$, what does Q_ϕ correspond to in Newton Mechanics? What is the dimensionality of $Q_\phi \, \delta\phi$?

(e) Based on this example, what do you learn about the difference between the orthogonal curvilinear coordinate system and the generalized coordinates?

Solution:

★ J02.09

A horizontal, circular table with a frictionless top surface is constrained to rotate about a vertical line through its center, with constant angular velocity ω_0. A peg is driven into the table top at a distance a from the center of the circle. A mass m slides freely on the top surface of the table, connected to the peg by a massless spring of force constance k and zero rest length. Take the s system to be an inertial system of Cartesian coordinates x, y with the origin at the center of the table top, and the q system to be rotating Carteisan coordinates x', y' defined so that $\vec{e}_x'$ defines a line passing through the peg. Ignore the z coordinate, and treat this problem as one with two degrees of freedom.The transformation between coordinates of the mass in the two systems is $x = x' \cos\omega_0 t - y' \sin\omega_0 t$, $y = x' \sin\omega_0 t + y' \cos\omega_0 t$



(a) Wirte $L(s, \dot{s}, t)$ in the s system and $L(q, \dot{q}, t)$ in the q system.

(b) Write H_s in the s system. Is it equal to $T + U$? Is it conserved?

(c) Write H_q in the q system. Is it equal to $T + U$? Is it conserved?

(d) What do you learn about the conserveness and generalized energy function?

Solution:

★ J02.12

A point particle of mass m and charge q moves near a very long wire carrying a current I. Choose the $\vec{e}_z$ axis along the wire in the direction of the current. In the region near the wire, the vector potential in terms of cylindrical polar coordinates ρ, ϕ, z is $\vec{A} = \frac{I}{2\pi c} \ln\left(\frac{\rho}{\rho_0}\right)\vec{e}_z$, where ρ_0 is some arbitrarily chosen ρ value. Assume the electric potential Φ to be zero.

(a) Write the Lagrangian $L = L(\rho, \phi, z, \dot{\rho}, \dot{\phi}, \dot{z}, t)$ for the particle, using cylindrical coordinates

(b) ind the generalized momenta p_ρ, p_ϕ, p_z

(c) Write three Lagrange equations, and show that ϕ and z are ignorable coordinates

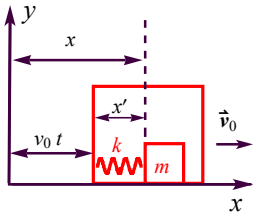
(d) Use the ϕ and z Lagrange equations to write expressions for $\dot{\phi}$ and $\dot{z}$ as a function of ρ and integration constants

(e) Write the generalized energy function. Is it conserved? Use it to express $\dot{\rho}^2$ as a function of ρ and some constants that can be determined at $t = 0$.

Solution:

★ 补充题

方盒相对于地面以 v_0 沿 x 向滑行，盒内质点 m 系于原长 l 弹性系数 k 的弹簧并可无摩擦滑动，在地面坐标系 Σ 写出质点的拉氏量 $L(x, \dot{x}, t)$。做广义坐标变换 $x = x(x' t)$，将 L 写成方盒坐标系 Σ' 坐标 x' 的函数 $L'(x', \dot{x}', t)$。若直接在 Σ' 系中写出拉格朗日量 L''，问 L' 和 L'' 是否相同？若不同请写出 $L'' - L'$。由 L' 和 L'' 得到的运动方程是否相同？由 L, L' 和 L'' 各自求出其广义能量函数 H, H' 和 H''，三者有何区别？各自的物理意义如何？



Solution: